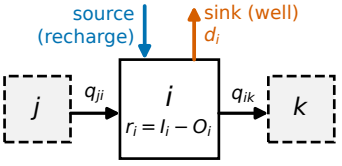
(a) one fully mixed cell

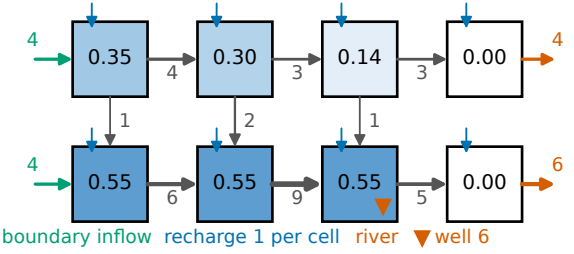


$I_i = \Sigma_j q_{ji} + \mathrm{Src}_i$

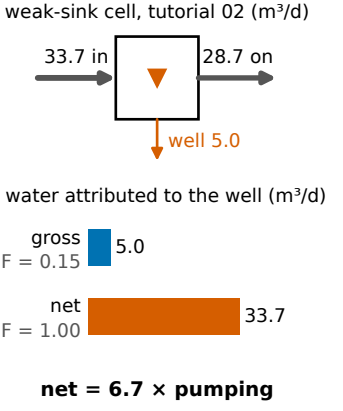
$O_i = \Sigma_k q_{ik} + \mathrm{Snk}_i$

$F_i = \Sigma_k (q_{ik}/O_i) F_k + d_i/O_i$

(b) backward fraction F to the well (gross)

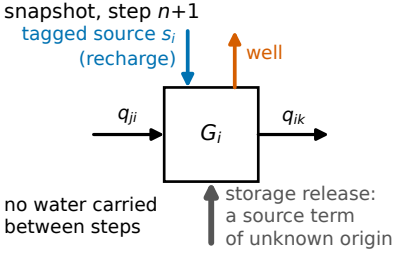


(c) gross vs net, weak sink

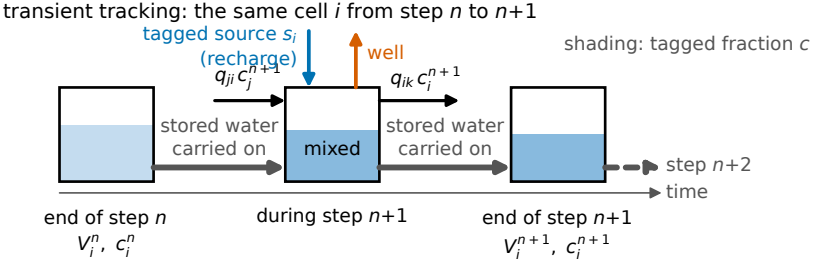


net = 6.7 × pumping

(d) transient models: snapshot vs transient tracking



$G_i = (\Sigma_j q_{ji} G_j + s_i) / I_i$



$$V_i^{n+1} c_i^{n+1} = V_i^n c_i^n + \Delta t\,(\Sigma_j q_{ji} c_j^{n+1} + s_i - O_i c_i^{n+1})$$